A Neural Network Approach to Predicting Bike Sharing Trip Duration

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Introduction

- Bike sharing systems like Chicago's Divvy provide a healthy, affordable, and eco-friendly way to travel
- Thousands of daily trips create large amounts of data about how people move around the city
- Predicting trip duration can help improve bike availability, station planning, and overall system efficiency

Research Objectives

- Predict bike trip duration using historical data
- Develop a **Neural Network (NN)** model to learn patterns in trip behavior
- Compare the neural network against a simple baseline model
- Evaluate whether the neural network improves prediction accuracy

Methodology

- **Dataset:** 1 million Chicago Divvy bike trips, shuffled and split into training, validation, and test sets (consistent split for fair evaluation)
- **Features:** time of day, day of week, month, and trip distance (used to capture temporal and spatial patterns)
- **Models: Linear Regression (baseline)** and a feedforward **Neural Network** implemented in PyTorch
- **Training:** neural network tuned using validation data with controlled hyperparameters; final model evaluated once on a held-out test set
- **Evaluation:** performance measured using RMSE, MAE, and MAPE to compare prediction accuracy

Discussion

Model Performance

- Neural network outperformed all baseline models (Linear, Ridge, Lasso)

Error Reduction

- The neural network consistently reduced prediction error across all metrics.
  - **RMSE**: 0.285 vs 0.333 (baselines)
  - **MAE**: 0.200 vs 0.248 (baselines)
  - **MAPE**: 22.36% vs 29.61% (baselines)

Key Insight

- Neural networks better capture complex, real-world patterns in trip data
- Leads to more accurate and reliable predictions than linear models

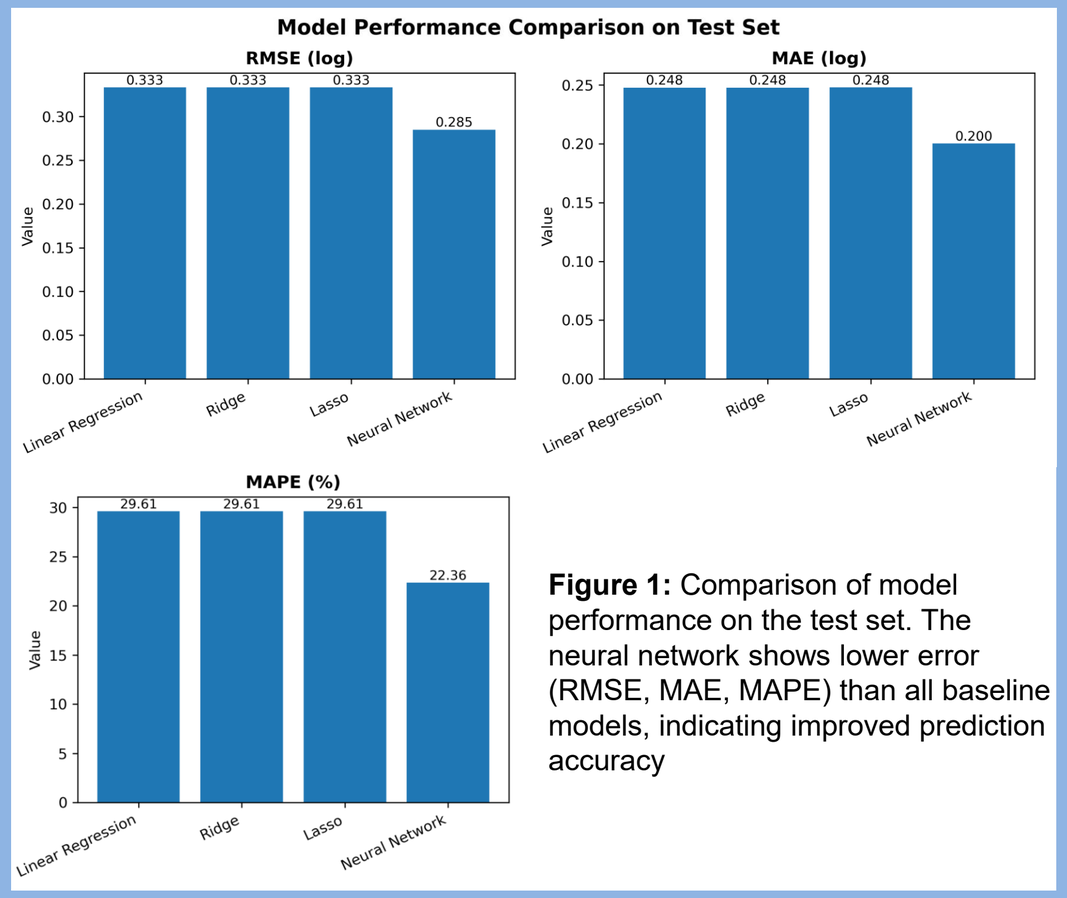

Figure 1: Comparison of model performance on the test set. The neural network shows lower error (RMSE, MAE, MAPE) than all baseline models, indicating improved prediction accuracy

Conclusion

- **Neural networks outperform simple baseline** models in predicting bike trip duration
- Neural networks **significantly reduce prediction error** across multiple metrics (RMSE, MAE, MAPE)
- The model successfully learns patterns from features such as time of day and trip distance
- Neural networks **capture complex, real-world relationships** that linear models cannot represent well
- **Proper tuning** (e.g., model size, learning rate) is important for achieving strong performance
- While neural networks improve accuracy, they require **more training time and careful setup** than simple models
- Overall, neural networks are effective tools for modeling large-scale urban transportation data

References

1. **Kruse et al. (2022)** *Computational Intelligence: A Methodological Introduction*, Springer.
2. **Yingwurenjian (n.d)** "Chicago Divvy Bicycle Sharing Data", Kaggle. https://www.kaggle.com/datasets/yingwurenjian/chicago-divvy-bicycle-sharing-data